

NDN Codebase Overview

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This Talk

- A quick overview of some major NDN software development
 - Mostly done at UCLA, Memphis, Florida International, NIST, Washington U and Arizona.
 - Many other great work done by the community not included.
- Recent progress and status

NFD: NDN Forwarding Daemon

- A C++ implementation of NDN packet forwarding
 - Content store, Pending Interest Table, FIB
 - Multiple forwarding strategies.
 - Expose its internals to applications through the NFD management protocol.
- Runs on multiple platforms
 - NFD-Android
 - OpenWRT/DD-WRT, Raspberry Pi, etc.
- Runs on the NDN Testbed
- ndn-tools
 - A collection of some basic NDN tools.

NFD: new release (v0.8) coming soon

- Complete the transition to packet format v0.3, including
 - New signed Interest format
 - New naming conventions
 - Dropped support of selectors, added support of HopLimit
- “Async behavior”
 - Buffering pending Interests that have no forwarding entry yet.
 - Forwarding them when a matching forwarding entry shows up.
- Improving out-of-the-box experience with Ethernet faces
- Revision of NFD Developer’s Guide

Other NDN Forwarders

- NDN-DPDK
 - Based on DPDK, multi-threaded, performance oriented, 100+ Gbps
 - Written in C and Go.
- YaNFD
 - Lightweight, portable, for the edge
 - multi-threaded with sharded data structures
 - Written in Go.
- NDN-Lite
 - Low-overhead forwarder targeted for embedded/constrained environments.
 - Integrated with a single application thread, provides
 - A package beyond a forwarder: pub/sub, schematized trust, access control etc.
 - Written in C.

Development Libraries

- ndn-cxx (C++)
- python-ndn
- NDNts (TypeScript)
- jNDN (Java)

Sync Protocols

- Sync is the transport protocol equivalent in NDN
 - Synchronize selected namespace and/or data among multiple participants
 - Different designs and implementations of Sync exist.
- PSync (Partial/Full Sync)
 - Using BF/IBF to represent names/states.
- StateVectorSync
 - Using a state vector to represent the states.

NDNCERT and Repo

- NDNCERT
 - Implement the NDN Certificate Management Protocol to automate the retrieval and management of NDN Certificates
 - Finalize v0.3 and its initial implementation.
- Repo: NDN Repository
 - Managed data repository
 - Re-designed protocol, Python implementation.

Evaluation Platforms

- ndnSIM: a widely used ns-3 based NDN simulator
 - Integrate native NFD code
 - Easy transition from simulation to real system
 - Allows simulation to use application code
- mini-NDN: a light-weight NDN emulator based on mininet.
 - Virtualized nodes that run NFD and other NDN software
 - A physical server can run an NDN network of tens of nodes.
- NDN Testbed
 - A global testbed with 30+ nodes
 - Runs NLSR as routing protocol and ASF as forwarding strategy.

Future Plan

- Accessibility/Usability
 - Auto-configuration, self-discovery, self-learning
- Security Support
 - Certificate management, schematized trust
- Support for other application needs